

Given: Due:	Further Maths (Statistics) HW 13	MARK SCHEME								
Chapter 1: Discrete Random Variables										
1) A fair blue die has faces numbered 1, 1, 3, 3, 5 and 5. The random variable B represents the score when the blue die is rolled.										
(a) Write down the probability distribution for B .		(2)								
(b) State the name of this probability distribution.		(1)								
(c) Write down the value of $E(B)$.		(1)								
A second die is red and the random variable R represents the score when the red die is rolled. The probability distribution of R is										
<table><tr><td>r</td><td>2</td><td>4</td><td>6</td></tr><tr><td>$P(R = r)$</td><td>$\frac{2}{3}$</td><td>$\frac{1}{6}$</td><td>$\frac{1}{6}$</td></tr></table>			r	2	4	6	$P(R = r)$	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$
r	2	4	6							
$P(R = r)$	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$							
(d) Find $E(R)$.		(2)								
(e) Find $\text{Var}(R)$.		(3)								
Tom invites Avisha to play a game with these dice. Tom spins a fair coin with one side labelled 2 and the other side labelled 5. When Avisha sees the number showing on the coin she then chooses one of the dice and rolls it. If the number showing on the die is <u>greater</u> than the number showing on the coin, Avisha wins, otherwise Tom wins. Avisha chooses the die which gives her the best chance of winning each time Tom spins the coin.										
(f) Find the probability that Avisha wins the game, stating clearly which die she should use in each case.		(4)								

Q1.

(a) was answered well and most gave the distribution in a table with values 1, 3 and 5 and probabilities of $\frac{1}{6}$ for each. Labelling each face as 1, 1, 3, 3, 5, 5 was quite common too, usually accompanied by probabilities of $\frac{1}{6}$ for each face, but some gave probabilities of $\frac{1}{3}$ for each face and others had values of 1, 2 and 3.

(b) was not answered well with many candidates only having one of the two words "discrete" and "uniform" in their answer. An answer of "discrete random distribution" was quite common but some candidates did not know the correct terms, describing it as an "equal" distribution.

Very few candidates used the symmetry of the distribution to "write down" the answer to (c) but they were still able to gain the mark by calculating $E(B)$. Some candidates could not answer part (d) correctly but in part (e) although a number found $E(R)^2 = \frac{34}{3}$ some thought this was $\text{Var}(R)$, others only subtracted 3, despite having previously quoted a correct formula, and some rounded their answer to 2.33 without first giving the exact value. In the final part of the question many candidates did not seem to understand what was required. Some drew complete tree diagrams but then did not realise that Avisha always chose the die that gave her the best chance of winning. Others identified which die she should choose without taking account of the probability of 0.5 for the outcomes of the spin of the coin. Nevertheless a small number of candidates did complete this part of the question successfully.

Question Number	Scheme	EXAM PAPERS PRACTICE	Marks								
(a)	<table border="1"> <tr> <td>b</td> <td>1</td> <td>3</td> <td>5</td> </tr> <tr> <td>$P(B=b)$</td> <td>$\frac{1}{3}$</td> <td>$\frac{1}{3}$</td> <td>$\frac{1}{3}$</td> </tr> </table> Also allow b values 1,1,3,3,5,5 and probabilities all $\frac{1}{6}$	b	1	3	5	$P(B=b)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$		B1 B1 (2)
b	1	3	5								
$P(B=b)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$								
(b)	Discrete Uniform {distribution}		B1 (1)								
(c)	$[E(B) =] 3$ (by symmetry)		B1 (1)								
(d)	$[E(R) =] 2 \times \frac{2}{3} + 4 \times \frac{1}{6} + 6 \times \frac{1}{6}$ $= 3$		M1 A1 (2)								
(e)	$[E(R^2) =] 2^2 \times \frac{2}{3} + 4^2 \times \frac{1}{6} + 6^2 \times \frac{1}{6} \quad \left[= \frac{34}{3} \right]$ $[Var(R) =] \frac{34}{3} - 3^2 = \frac{7}{3}$ (or any exact equivalent. NB 2.33 is A0)		M1 dM1, A1 (3)								
(f)	Coin lands on 2, choose blue die; coin lands on 5 choose red die $P(\text{Avisha wins}) = \frac{1}{2} \times \left(\frac{1}{3} + \frac{1}{3} \right) + \frac{1}{2} \times \frac{1}{6}$ $= \frac{5}{12}$ (allow awrt 0.417)		B2/1/0 M1 A1 (4)								
Notes			13								
(a)	1^{st} B1 for correctly identifying values of b as 1, 3, 5 or 1,1,3,3,5,5 2^{nd} B1 for probabilities all $= \frac{1}{3}$ or exact equivalent (or of course 6 cases of $\frac{1}{6}$) Any correct probability distribution or probability function is 2/2. Must be in part (a)										
(b)	B1 for "Discrete Uniform". Both words required.										
(c)	B1 for answer of 3 o.e. Accept $E(X) = 3$										
(d)	M1 for an attempt at correct formula. At least 2 correct products seen. If later divide by $n (\neq 1)$ M0 A1 for an answer of 3. Correct answer only scores both marks.										
(e)	1^{st} M1 for a correct attempt at $E(R^2)$. At least 2 correct products seen. Condone $Var(R) = \text{etc}$ May be implied by sight of $\frac{34}{3}$ or 11.3 or better. 2^{nd} dM1 Dep. on 1^{st} M1 for clear attempt at $E(R^2) - [E(R)]^2$ Must see their values <u>used</u> . NB $Var(R) = E(R^2) - [E(R)]^2 = \frac{34}{3} - "3"$ is M1M0A0 since do not <u>use</u> their $[E(R)]^2$										
(f)	B2/1/0 Both correct B1B1, one correct B1B0. Do not use B0B1 [e.g. always red or RR is B1B0] NB Allow other descriptions of the die e.g. 1^{st} or fair for blue, 2^{nd} for red if they are clear. M1 for evaluating correct probabilities i.e. only $\frac{1}{3}, \frac{1}{12}$ seen <u>or</u> if incorrect choice made: M1 for an answer of : if choose RR ($\frac{1}{3}$), if choose BB ($\frac{1}{3}$), if choose RB ($\frac{1}{6}$) NB $\frac{5}{12}$ as answer scores M1A1. Need to see choices of die stated for B marks.										

- (a) 1st B1 for correctly identifying values of b as 1, 3, 5 or 1,1,3,3,5,5
 2nd B1 for probabilities all $= \frac{1}{3}$ or exact equivalent (or of course 6 cases of $\frac{1}{6}$)
 Any correct probability distribution or probability function is 2/2. Must be in part (a)
- (b) B1 for "Discrete Uniform". Both words required.
- (c) B1 for answer of 3 o.e. Accept $E(X) = 3$
- (d) M1 for an attempt at correct formula. At least 2 correct products seen. If later divide by $n (\neq 1)$ M0
 A1 for an answer of 3. Correct answer only scores both marks.
- (e) 1st M1 for a correct attempt at $E(R^2)$. At least 2 correct products seen. Condone $Var(R) = \text{etc}$
 May be implied by sight of $\frac{34}{3}$ or 11.3 or better.
 2nd dM1 Dep. on 1st M1 for clear attempt at $E(R^2) - [E(R)]^2$ Must see their values used.
 NB $Var(R) = E(R^2) - [E(R)]^2 = \frac{34}{3} - 3^2$ is M1M0A0 since do not use their $[E(R)]^2$
- (f) B2/1/0 Both correct B1B1, one correct B1B0. Do not use B0B1 [e.g. always red or RR is B1B0]
 NB Allow other descriptions of the die e.g. 1st or fair for blue, 2nd for red if they are clear.
 M1 for evaluating correct probabilities i.e. only $\frac{1}{3}, \frac{1}{12}$ seen or if incorrect choice made:
 M1 for an answer of : if choose RR ($\frac{1}{4}$), if choose BB ($\frac{1}{3}$), if choose RB ($\frac{1}{6}$)
 NB $\frac{5}{12}$ as answer scores M1A1. Need to see choices of die stated for B marks.

2) The score S when a spinner is spun has the following probability distribution.

s	0	1	2	4	5
$P(S=s)$	0.2	0.2	0.1	0.3	0.2

- (a) Find $E(S)$. (2)
- (b) Show that $E(S^2) = 10.4$ (2)
- (c) Hence find $Var(S)$. (2)
- (d) Find (4)
- (i) $E(5S - 3)$,
- (ii) $Var(5S - 3)$.
- (e) Find $P(5S - 3 > S + 3)$ (3)
- The spinner is spun twice.
 The score from the first spin is S_1 and the score from the second spin is S_2
 The random variables S_1 and S_2 are independent and the random variable $X = S_1 \times S_2$
- (f) Show that $P(\{S_1 = 1\} \cap X < 5) = 0.16$ (2)
- (g) Find $P(X < 5)$. (3)

Parts (a), (b), (c) and (d) were answered very well and most candidates scored well here. In part (e) most

knew they had to solve the inequality and arrived at $S > 1.5$ but some misinterpreted this and gave $P(S > 1)$ rather than $P(S > 2)$. The final 2 parts caused problems for many candidates. A popular approach was to calculate a sample space showing all 25 products of S_1 and S_2 but they often thought that these 25 cases were all equally likely. Unfortunately this false assumption would lead to the correct answer in part (f), but not of course the available marks, and many candidates never properly considered the correct distribution of X . Some candidates did list the 15 required cases and were able to obtain the correct probability but such cases were rare.

Question	Scheme	Marks
(a)	$E(S) = 0 + 1 \times 0.2 + 2 \times 0.1 + 4 \times 0.3 + 5 \times 0.2 = [0.2 + 0.2 + 1.2 + 1.0]$ <u>2.6</u>	M1 A1 (2)
(b)	$E(S^2) = 0 + 1 \times 0.2 + 2^2 \times 0.1 + 4^2 \times 0.3 + 5^2 \times 0.2$ or $0.2 + 0.4 + 4.8 + 5$ <u>10.4</u> (*)	M1 A1 also (2)
(c)	$\text{Var}(S) = 10.4 - ("2.6")^2$ <u>3.64</u> or $\frac{91}{25}$ (o.e.)	M1 A1 (2)
(d)(i)	$5E(S) - 3 = 5 \times "2.6" - 3$, = <u>10</u>	M1, A1
(ii)	$5^2 \text{Var}(S) = 25 \times 3.64$, = <u>91</u>	M1, A1 (4)
(e)	$5S - 3 > S + 3 \Rightarrow 4S > 6$ or $S > 1.5$, so need $P(S \geq 2)$ $P(S \geq 2) = \underline{0.6}$	M1, A1 A1 (3)
(f)	$P(S_1 = 1) \times P(S_2 \leq 4) = 0.2 \times 0.8 = 0.16$ (*)	M1, A1 also (2)
(g)	$P(S_1 = 2) \times P(S_2 \leq 2) = 0.1 \times 0.5 = 0.05$ $P(S_1 = 4) \times P(S_2 \leq 1) = 0.3 \times 0.4 = 0.12$ Full method – all cases listed $P(S_1 = 5) \times P(S_2 = 0) = 0.2 \times 0.2 = 0.04$ all correct products $P(S_1 = 0) \times P(S_2 = \text{any value}) = 0.2 \times 1 = 0.20$ <u>0.57</u>	M1 A1 A1 (3) [18]
Notes		
(a)	M1 for an attempt at $\sum xP(X=x)$, at least 2 non-zero terms seen. Correct answer 2/2 A1 for 2.6 or any exact equivalent	
(b)	M1 for a correct attempt, at least 3 non-zero terms seen A1 also for 10.4 provided M1 is scored and no incorrect working seen	
(c)	M1 for $10.4 - \mu^2$, ft their μ . Must see their value of μ squared (A1 for 3.64 or any exact equiv.)	
(d)(i)	M1 for a correct expression using their 2.6 (A1 for 10)	
(ii)	M1 for $25 \times \text{Var}(S)$ - ft their $\text{Var}(S)$ (A1 for 91)	
(e)	M1 for solving the inequality as far as $pS > q$ where one of p or q are correct 1 st A1 for $P(S \geq 2)$ 2 nd A1 for 0.6 (provided $S > 1.5$ was obtained). Ans only of 0.6 scores 3/3	
(f)	A table showing all 25 cases can only score M1 in (g) if the correct cases are indicated. M1 for using independence (so multiplying) and attempting $P(S_2 \leq 4)$ e.g. $0.2 \times (0.2 + 0.2 + 0.1 + 0.3)$ or $0.04 + 0.04 + 0.02 + 0.06$ score M1 BUT $\frac{4}{25}$ (not from 0.2×0.8) is M0A0 A1 also for a fully correct explanation leading to 0.16. Must come from 0.2×0.8 not $\frac{4}{25}$	
(g)	M1 for all cases for S_1 or all 15 cases for X 1 st A1 for all correct probability products for S_1 or X 2 nd A1 for 0.57 Correct answer scores 3/3. Probabilities out of 25 score A0A0	

3) The random variable X has probability distribution

x	1	3	5	7	9
$P(X=x)$	0.2	p	0.2	q	0.15

(a) Given that $E(X) = 4.5$, write down two equations involving p and q .

(3)

Find

(b) the value of p and the value of q ,

(3)

(c) $P(4 < X \leq 7)$.

(2)

Given that $E(X^2) = 27.4$, find

(d) $\text{Var}(X)$,

(2)

(e) $E(19 - 4X)$,

(1)

(f) $\text{Var}(19 - 4X)$.

(2)

A sizeable minority of candidates did not attempt this question, but, when attempted, this question was well done with many candidates picking up most or all the marks. Rather surprisingly in part (a) the equation missing was $p+q=0.45$ and a few candidates divided one side of their second equation by 5. In part (b) nearly all of those who had two correct equations for part (a) were able to solve them simultaneously. In part (c) a substantial number of candidates were unable to make a successful attempt at this part of the question with many omitting it entirely. There were a large number of accurate solutions to part (d) with most of those making an error failing to use the given values. A number of candidates reworked $E(X^2)$ for part (e) even though it was given. There were some mistakes in part (f) but most candidates used 16 correctly, but some multiplied $E(X)$ instead of $\text{Var}(X)$.

Question Number	Scheme	Marks
(a)	$p+q=0.45$ $\sum xP(X=x)=4.5$ $3p+7q=1.95$	B1 M1 A1 (3)
(b)	Attempt to solve equations in (a) $q=0.15$ $p=0.30$	M1 A1 A1 (3)
(c)	$P(4 < X < 7) = P(5) + P(7)$ $= 0.2 + q = 0.35$	M1 A1 $\int$ (2)
(d)	$\text{Var}(X) = E(X^2) - [E(X)]^2 = 27.4 - 4.5^2$ $= 7.15$	M1 A1 (2)
(e)	$E(19 - 4X) = 19 - 4 \times 4.5 = 1$	B1 (1)
(f)	$\text{Var}(19 - 4X) = 16\text{Var}(X)$ $= 16 \times 7.15 = 114.4$	M1 A1 (2)
		Total 13 marks

Notes:	
(a)	$0.55 + p + q = 1$ award B1. Not seen award B0. $0.2 + 3p + 1 + 7q + 1.35 = 4.5$ or equivalent award M1A1 $3p + 7q + k = 4.5$ award M1.
(b)	Attempt to solve must involve 2 linear equations in 2 unknowns Correct answers only for accuracy. Correct answers with no working award 3/3
(c)	Follow through accuracy mark for their q , $0 < q < 0.8$
(d)	Attempt to substitute <u>given</u> values <u>only</u> into correct formula for M1. 7.15 only for A1 7.15 seen award 2/2
(f)	Accept 'invisible brackets' i.e. $-4^2 \text{Var}(X)$ provided answer positive. Anything that rounds to 114 for A1.

- 4) A fair six-sided black die has faces numbered 1, 2, 2, 3, 3 and 4
 The random variable B represents the score when the black die is rolled.
 (a) Write down the value of $E(B)$

(1)

A white die has 6 faces numbered 1, 1, 2, 4, 5 and c where $c > 5$
 The discrete random variable W represents the score when the white die is rolled and has probability distribution given by

w	1	2	4	5	c
$P(W=w)$	$a+b$	a	0.3	a	b

Greg and Nilaya play a game with these dice.

Greg throws the black die and Nilaya throws the white die. Greg wins the game if he scores at least two more than Nilaya, otherwise Greg loses.

The probability of Greg winning the game is $\frac{1}{6}$

- (b) Find the value of a and the value of b
 Show your working clearly.

The random variable $X = 2W - 5$

Given that $E(X) = 2.6$

(c) find the exact value of $\text{Var}(X)$

The majority of students were able to gain the mark for part (a).

Part (b) proved to be a challenge for all but the most able. Those who wrote down the ways in which Greg could win fared best as they were then able to attempt to find the probability in terms of a and b however, this was rarely seen. Most students tried to just write down the probability of Greg winning straight away with little success. However, they usually managed to use the fact that the total of the probabilities must equal one to gain the mark for $3a + 2b = 0.7$

In part (c) students used a mixture of possible methods. The majority of students knew the correct formula for the variance but did not check it was all in W or X . All too often they used $E(X^2) - [E(W)]^2$. Another common error was to try to find $2E(X^2)$ using $E(2W - 5)$.

Question	Scheme	Marks	AOs								
(a)	2.5	B1	1.1b								
		(1)									
(b)	A complete strategy to find a value for a and a value for b .	dM1	3.1b								
	<table><tr><td>Greg</td><td>Nilaya</td></tr><tr><td>4</td><td>1</td></tr><tr><td>4</td><td>2</td></tr><tr><td>3</td><td>1</td></tr></table>	Greg	Nilaya	4	1	4	2	3	1	M1	2.1
Greg	Nilaya										
4	1										
4	2										
3	1										
	$\frac{1}{6}(2a+b) + \frac{1}{3}(a+b) = \frac{1}{6}$ oe $4a+3b=1$ oe	M1	1.1b								
	$3a+2b=0.7$ oe	M1	1.1b								
	$a=0.1 \quad b=0.2$	A1	1.1b								
		(5)									
(c)	$2E(W) - 5 = 2.6$	M1	3.1a								
	$E(W) = 3.8$										
	$8a+b+cb+1.2 [=3.8]$ or " 0.3 " + $2 \times$ " 0.1 " + 4×0.3 + $5 \times$ " 0.1 " + " 0.2 " $c =$ [" 3.8 "]	M1	1.1b								
	$c = \frac{2.6 - 8"a" - "b"}{"b"}$										
	$c = 8$	A1	1.1b								
	$E(W^2) = 30a + c^2b + b + 4.8$ or " 0.3 " + $4 \times$ " 0.1 " + 16×0.3 + $25 \times$ " 0.1 " + " 8 " $\times$ 0.2	M1	1.1b								
	$E(W^2) = 20.8$										
	$\text{Var}(W) = "20.8" - "3.8"$ $\times$ 2 or 6.36	M1	1.1a								
	$\text{Var}(X) = 25.44$	A1ft	1.2								
		(6)									
(12 marks)											

Notes

EXAM PAPERS PRACTICE

(a) B1: $\frac{5}{2}$ or 2.5

(b) dM1: Dependent on 3rd and 4th Method marks being awarded.

For a complete strategy to find a value of a and a value of b .

Need 2 independent equations in a and b , one equation must be $\text{prob} = 1/6$ and the other

$3a + 2b = 0.7$ oe and an attempt to solve. For an attempt we require a method to eliminate one variable leading to a value for a and b , or correct values.

M1: For using the given contextual information to list 3 different combinations for Greg to win.

Implied by $4a + 3b = 1$ oe

M1: For using $P(g) \times P(n)$ for each combination identified as a win for Greg $= \frac{1}{6}$

It must be a linear equation in a and b with 2/3 terms on the LHS, at least one of which must be correct and equal to $1/6$

M1: For use of $\sum P(W = w) = 1$

A1: For both values correct

(c) M1: For translating the given mathematical context into an expression for $E(W)$

May be implied by a correct equation or $E(W) = 3.8$

M1: For use of $\sum wP(W = w) [= 3.8]$ If algebraic then at least 2 terms must be correct, if numerical at least 3 terms correct ft their values of a and b . This must be seen in part (c)

[NB: $16a + 2b + 2cb + 2.4 - 5 = 2.6$ oe would get M1M1]

A1: cao

M1: For use of $\sum w^2P(W = w)$. If algebraic then at least 2 terms must be correct, if numerical at least 3 terms correct ft their values of a and b .

M1: For use of $\text{Var}(W) = E(W^2) - [E(W)]^2$

A1ft: $4 \times \text{"their Var}(W)"$ ft their $\text{Var}(W)$ provided a, b and $\text{Var}(W)$ are > 0 and $c > 5$

Alternative for (c). Allow a mix of methods

(c)	$[E(X) =] -3 \times (a+b) - 1 \times a + 0.9 + 5 \times a + (2c-5) \times b$	M1
	$-3 \times (a+b) - 1 \times a + 0.9 + 5 \times a + (2c-5) \times b = 2.6$	M1
	$c = 8$	A1
	Values of X -3, -1, 3, 5, $2c-5$	M1
	$E(X^2) = 9 \times (a+b) + 1 \times a + 2.7 + 25 \times a + (2c-5)^2 \times b$	M1
	$= 32.2$	
	$\text{Var}(X) = "32.2" - 2.6^2$	A1ft
	$= 25.44$	
Notes for alternative		
	M1: allow with their a, b and X values	
	M1: allow with their a, b and X values	
	A1: cao	
	M1: at least 3 correct	
	M1: allow with their c, a, b and X values	
	A1ft: ft their $E(X^2)$ provided a and b are > 0 and $c > 5$	